

AI-AUGMENTED BACKEND DEVELOPMENT

AI BlogNest API

2. Requirement Analysis Phase

Functional, non-functional, software, hardware, user, and AI requirements

1. Requirement Analysis Objective

The requirement analysis phase converts the initial idea into functional, technical, security, and infrastructure requirements. The sample specifies the software and hardware environment and describes the required backend capabilities. ■filecite■turn0file0■L49-L63■

2. Functional Requirements

- User registration and authentication.
- JWT-based protected API access.
- Role-based authorization for Admin, Editor, Author, and Reader.
- Create, read, update, and delete blog content according to permissions.
- Blog lifecycle management including Draft, Pending Approval, Scheduled, and Published states.
- Category and tag management.
- Comment creation and moderation.
- Media metadata/path registration.
- Advanced blog search and filtering.
- AI blog generation through a dedicated service endpoint.
- AI blog summarization through a dedicated service endpoint.

3. Non-Functional Requirements

- Security: password hashing, JWT validation, request validation and sanitization.
- Reliability: centralized error handling and structured logging.
- Performance: efficient MongoDB access through Mongoose.
- Scalability: modular service architecture and separation of concerns.
- Maintainability: clear model, controller, route, middleware, and service responsibilities.
- Usability for API consumers: predictable JSON responses and standard HTTP methods.

4. Software Requirements

- Windows 10/11, macOS, or Linux.
- Node.js v16 or above.
- npm v8 or above.

- Express.js.
- MongoDB.
- Postman or an equivalent API verification tool.
- Visual Studio Code or another code editor.

5. Hardware Requirements

- Intel Core i5 8th Gen or AMD Ryzen 5 or equivalent.
- At least 8 GB RAM; 16 GB is recommended for concurrent database and development processes.
- At least 1 GB available development storage.

6. User Requirements

The system supports Admin, Editor, Author, and Reader roles. Admin controls users and system-level operations; Editors handle editorial workflows; Authors create posts associated with their accounts; Readers consume published posts and can comment. ■filecite■turn0file0■L126-L135■

7. AI Requirements

The AI layer must accept meaningful thematic input, apply appropriate context and prompt structure, return generated or summarized content, and remain integrated with the backend's authentication and request-processing pipeline. The sample identifies dedicated AI routes such as /api/ai/generate-blog and /api/ai/summarize. ■filecite■turn0file0■L79-L85■